

Response Surface Methodology

Lin Wang

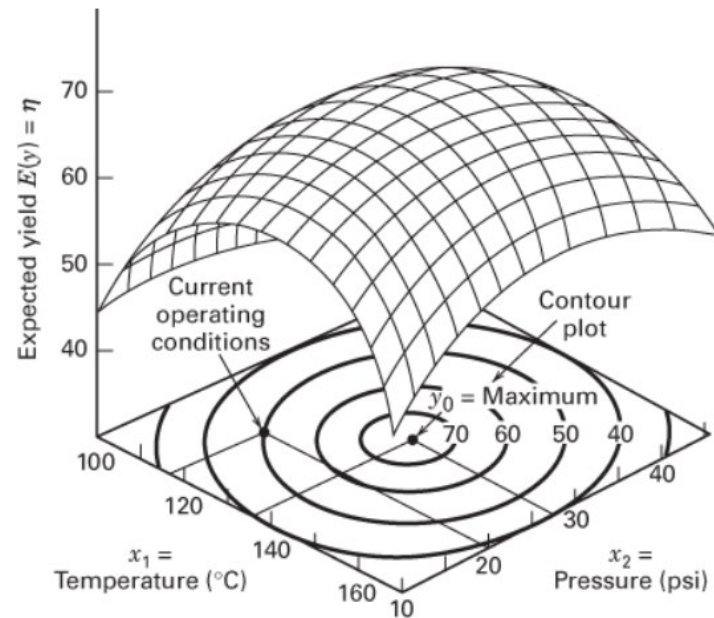
Response Surface Methodology

- **Objective** of the Response Surface Methodology is **optimization**, that is, to find out the optimal settings of several factors so that the response is optimized.
- Applications of RSM span many industrial and business settings

Response Surface Methodology

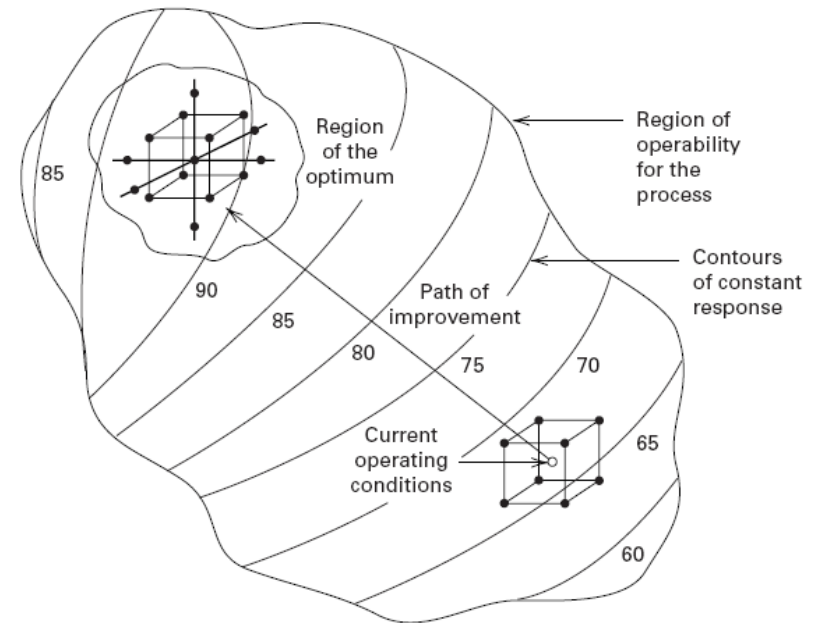
Example: an engineer wishes to find the levels of two factors x_1 and x_2 to maximize the yield y .

Response surface
 $y = f(\mathbf{x})$



RSM is a Sequential Procedure

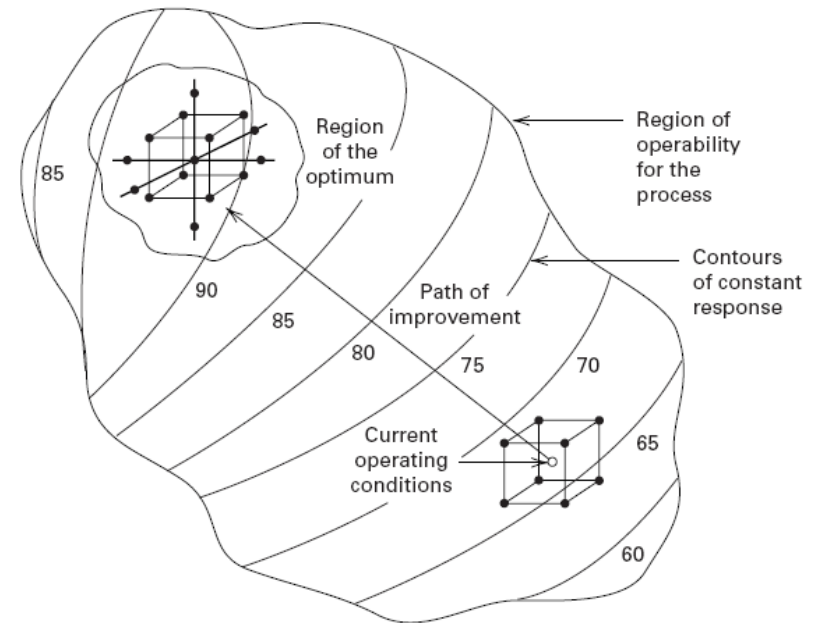
- When we are at a point on the response surface that is remote from the optimum, such as the current operating conditions in Figure 11.3, there is little curvature in the system and the first-order model will be appropriate. Our objective here is to lead the experimenter rapidly and efficiently along a path of improvement toward the general vicinity of the optimum.
- Once the region of the optimum has been found, a more elaborate model, such as the second-order model, may be employed, and an analysis may be performed to locate the optimum.



■ FIGURE 11.3 The sequential nature of RSM

RSM is a Sequential Procedure

- The analysis of a response surface can be thought of as “climbing a hill,” where the top of the hill represents the point of maximum response.
- If the true optimum is a point of minimum response, then we may think of “descending into a valley.”



■ FIGURE 11.3 The sequential nature of RSM

Steps in RSM

1. Factor screening
2. Conduct local experiment.
3. Find a suitable approximation for $y = f(\mathbf{x})$ using LS {maybe a low – order polynomial}
4. Perform curvature test. If curvature is not found, move towards the region of the optimum (The Method of Steepest Ascent)
5. Repeat Steps 2—4 until curvature is found.
6. When curvature is found, find a new approximation for $y = f(\mathbf{x})$ {generally a higher order polynomial} and perform the “Response Surface Analysis”

Response Surface Models

- Low-order polynomial

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

- Higher-order polynomial

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \varepsilon$$

Example 11.1: An Example of Steepest Ascent

A chemical engineer is interested in determining the operating conditions that maximize the yield of a process. Two controllable variables

- reaction time: current setting is 35 minutes
- reaction temperature: current setting is 155°F

The engineer first explores (30, 40) minutes for reaction time and (150, 160) for temperature.

If ξ_1 denotes the natural variable time and ξ_2 denotes the natural variable temperature, then the coded variables are

$$x_1 = \frac{\xi_1 - 35}{5} \quad \text{and} \quad x_2 = \frac{\xi_2 - 155}{5}$$

Example 11.1: An Example of Steepest Ascent

TABLE 11.1

Process Data for Fitting the First-Order Model

Natural Variables		Coded Variables		Response
ξ_1	ξ_2	x_1	x_2	y
30	150	-1	-1	39.3
30	160	-1	1	40.0
40	150	1	-1	40.9
40	160	1	1	41.5
35	155	0	0	40.3
35	155	0	0	40.5
35	155	0	0	40.7
35	155	0	0	40.2
35	155	0	0	40.6

- The design is a 2^2 factorial design augmented by five center points.
- Center points are to study the current operating conditions
- Replicates at the center are used to estimate the experimental error and to allow for checking the adequacy of the first-order model.

A first-order model may be fit to these data by least squares:

$$\hat{y} = 40.44 + 0.775x_1 + 0.325x_2$$

The adequacy of the first-order model should be investigated. The 2^2 design with center points allows the experimenter to do the following:

1. Obtain an estimate of error.
2. Check for interactions (cross-product terms) in the model.
3. Check for quadratic effects (curvature).

1. Obtain an estimate of error.

$$\hat{\sigma}^2 = \frac{(40.3)^2 + (40.5)^2 + (40.7)^2 + (40.2)^2 + (40.6)^2 - (202.3)^2/5}{4} = 0.0430$$

2. Check for interactions (cross-product terms) in the model.

$$\hat{\beta}_{12} = \frac{1}{4}[(1 \times 39.3) + (1 \times 41.5) + (-1 \times 40.0) + (-1 \times 40.9)] = \frac{1}{4}(-0.1) = -0.025$$

$$SS_{\text{Interaction}} = \frac{(-0.1)^2}{4} = 0.0025$$

$$F = \frac{SS_{\text{Interaction}}}{\hat{\sigma}^2} = \frac{0.0025}{0.0430} = 0.058$$

P-value=0.8215, indicating that interaction is negligible.

3. Check for quadratic effects (curvature).

$\bar{y}_F - \bar{y}_C$ is an estimate of $\beta_{11} + \beta_{22}$.

$$\hat{\beta}_{11} + \hat{\beta}_{22} = \bar{y}_F - \bar{y}_C = 40.425 - 40.46 = -0.035$$

$$SS_{\text{Pure Quadratic}} = \frac{n_F n_C (\bar{y}_F - \bar{y}_C)^2}{n_F + n_C} = \frac{(4)(5)(-0.035)^2}{4+5} = 0.0027$$

$$F = \frac{SS_{\text{Pure Quadratic}}}{\hat{\sigma}_2^2} = \frac{0.0027}{0.0430} = 0.063$$

TABLE 11.2

Analysis of Variance for the First-Order Model

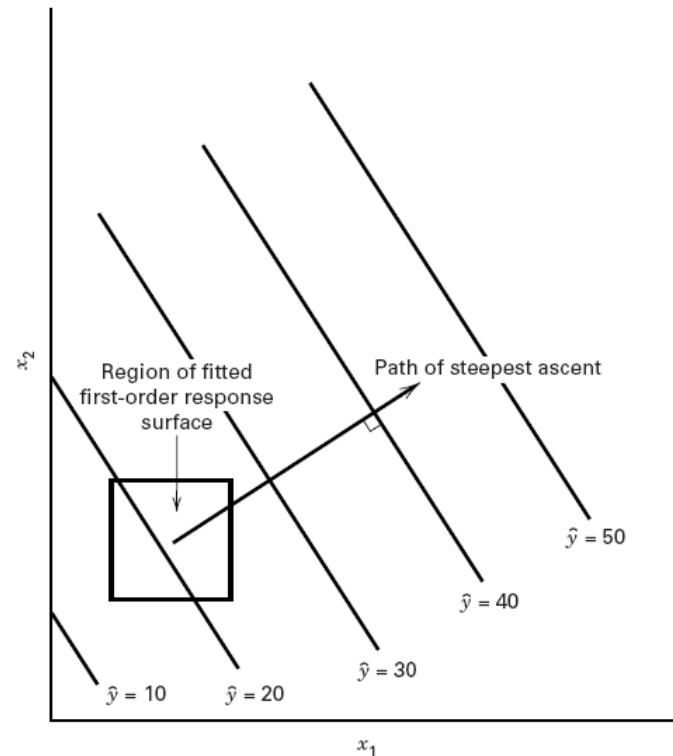
Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P-Value
Model (β_1, β_2)	2.8250	2	1.4125	47.83	0.0002
Residual	0.1772	6			
(Interaction)	(0.0025)	1	0.0025	0.058	0.8215
(Pure quadratic)	(0.0027)	1	0.0027	0.063	0.8142
(Pure error)	(0.1720)	4	0.0430		
Total	3.0022	8			

The Method of Steepest Ascent

- A procedure for moving sequentially from an initial “guess” towards region of the optimum
- Based on the fitted first-order model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- Steepest ascent (or descent) is a gradient procedure



$$\hat{y} = 40.44 + 0.775x_1 + 0.325x_2$$

- To move away from the design center—the point ($x_1 = 0, x_2 = 0$)—along the path of steepest ascent, we would move 0.775 units in the x_1 direction for every 0.325 units in the x_2 direction.
- Thus, the path of steepest ascent passes through the point ($x_1 = 0, x_2 = 0$) and has a slope $0.325/0.775$.
- Decide a step size: $\Delta x_1 = 1$ (that is, 5 minutes of reaction time)
- Therefore, the steps along the path of steepest ascent are $\Delta x_1 = 1$ and $\Delta x_2 = (0.325/0.775) = 0.42$.